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EVALUATING PROSODIC FEATURES OF SYNTHETIC SPEECH

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ABSTRACT

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Speech synthesis has developed rapidly due to artificial intelligence. Modern neural-based text-to-speech (TTS) systems can generate intelligible and natural sounding speech. The naturalness and intelligibility of synthetic speech can be expressed with prosodic properties. State-of-the-art TTS models are capable of explicitly learn prosodic features. However, the generation of prosodically rich speech is a challenging task.

The goal of this thesis is to evaluate how well synthetic speech inherits prosodic features of natural speech it is trained on. The evaluation was conducted by comparing the prosodic features of real and synthetic speech. Compared features include fundamental frequency (F0), speech rate, energy, and spectral tilt.

The data used in the experiments consists mainly of the LJSpeech dataset. The LJSpeech consists of audiobooks from LibriVox audiobook domain. Recordings for natural speech were collected from the LibriVox, which were not included in LJSpeech dataset. Synthetic speech was generated with FastPitch TTS model and HiFi-GAN voice coder, which were trained with LJSpeech. FastPitch uses pitch and duration predictions for the generation of mel-spectrogram. Overall, there were six audiobook chapters with a total length of 2.5 hours. Features were extracted using Python libraries.

The results show that there is difference between prosodic distributions of real and synthetic speech. The TTS system was able to produce high quality synthetic speech. Large statistical difference was found for F0 and speech rate. Synthetic speech had significantly higher speech rate with less variation in pitch contour. The least statistical different features were energy and spectral tilt.

Keywords: Text-to-speech, synthetic speech evaluation, prosodic features

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TIIVISTELMÄ

Arttu Raatikainen: Synteettisen puheen prosodisten ominaisuuksien evaluointi
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Puhesynteesi on kehittynyt vauhdilla tekoälyn kehittymisen myötä ja modernit järjestelmät käyttävät koneoppimisen menetelmiä. Tekstistä puheeksi -synteessillä (engl. text-to-speech, TTS) tarkoitetaan järjestelmää, joka muuttaa kirjoitetun tekstin ääneksi. Nykyaikaiset järjestelmät kykenevät tuottamaan ymmärrettävää ja luonnollisen kuuloista puhetta. Luonnollisuutta voidaan havainnollistaa prosodisten eli suprasegmentaalisten ominaisuuksien avulla. Prosodisia ominaisuuksia voidaan tarkastella akustisesti äänisignaalista. Vaikka TTS-järjestelmät ovat kehittyneet vauhdilla, niillä on haasteita tuottaa prosodisesti rikasta puhetta.

Tämän työn tarkoitus on evaluoida kuinka hyvin puhesynteesi kykenee perimään prosodiset ominaisuudet puhujalta, jolla se on koulutettu. Evaluointi on tehty vertailemalla ihmisäänen ja synteettisen puheen prosodisia ominaisuuksia. Ominaisuudet ovat fundamentaalitaajuus (F0), puhenopeus, intensiteetti ja äänisignaalin taajuusspektrin kaltevuus.

Tutkimuksen ääninäytteet koostuivat pääosin LJSpeech-datasetistä, joka on luotu äänikirjojen nauhoituksista. LJSpeech on laajasti käytetty datasetti TTS-järjestelmien kouluttamiseen ja sitä käytettiin tässä työssä FastPitch TTS-mallin ja HiFi-GAN vokooderin kouluttamiseen. Ihmisääni kerättiin LibriVox-äänikirjakokoelmasta, jotka eivät sisälly LJSpeech-datasettiin. Vertailtavia nauhoituksia oli yhteensä kuusi kappaletta, joiden yhteiskesto on noin 2,5 tuntia. Akustiset ominaisuudet laskettiin äänisignaalista käyttäen Python kirjastoja.

Työssä onnistuttiin luomaan korkealaatuista synteettistä puhetta. Tulokset osoittavat prosodisten ominaisuuksien eroavaisuudet ihmisäänen ja synteettisen puheen välillä. Tilastollisesti suurimmat erot löydettiin F0:lle ja puhenopeudelle. Synteettinen puhe sisältää selkeästi nopeatempoisempaa puhetta vähemmällä taajuusvaihtelulla. Tulokset osoittavat TTS-järjestelmän haasteet taajuuden ja puhekeston generoimisessa. Pienimmät erot tilastollisesti olivat intensiteetin ja taajuusspektrin kaltevuuden välillä.

Avainsanat: Puhesynteesi, synteettisen puheen evaluointi, prosodiset ominaisuudet

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1. INTRODUCTION

Speech synthesis, also known as text-to-speech (TTS), is technology to produce human speech from text input. In recent years, TTS systems have rapidly advanced through the development of artificial intelligence. State-of-the-art TTS systems can produce speech which quality is close to natural speech. Speech synthesis is able to achieve high naturalness for neutral-style speech, although the generation of prosodically rich speech is still a challenging task (Zhang et al. 2023).

Prosodic properties of speech are suprasegmental properties including melody, loudness, tempo, and spectral quality of speech. Prosody helps to express linguistic meanings, emotions, and speaker attitudes in speech. State-of-the-art TTS models are capable of explicitly learn prosodic features. For example, FastPitch is able to model melodic and temporal features of segmental units (Łańcucki 2021).

Digital devices use TTS to communicate with people. Natural sounding speech makes communication more efficient and convenient. Quality of synthetic speech produced by TTS systems is commonly evaluated by the human ear. Gathering human evaluators and setting up tests takes time and resources. Statistical analysis on measured distributions could make evaluating automated.

The goal of this study is to evaluate how well synthetic speech inherits prosodic features of natural speech it is trained on. The research is conducted by comparing synthetic speech and real speech produced by the same speaker whose speech was used for TTS training. Measured features include speech rate, energy, pitch, and spectral tilt.

Chapter 2. presents theory on TTS systems, prosodic features of speech and evaluation metrics. Chapter 3. introduces related work on TTS speech quality evaluation. The results of experiments are presented in Chapter 4., including discussion and feature extraction methods. Chapter 5. summarizes the results and discusses future work and improvements.

2. THEORY

2.1 Text-to-speech synthesis

The base idea of TTS is to convert written text to speech. Neural network (NN) based systems have become dominant methods in the field of speech synthesis (Xu et al. 2021). Modern TTS systems are able to generate more natural and intelligible speech with high quality.

Van Den Oord et al. (2016) introduced the WaveNet, which was the first model using a deep neural network architecture for raw audio generation (Triantafyllopoulos et al. 2023). WaveNet generates the audio signal directly from the linguistic features. There are generally three main components in TTS pipeline: text analysis, acoustic model, and voice coder (vocoder). These components might be trained and used separately or joined together into one, so called “end-to-end” system. Wang et al. (2016) introduced an end-to-end TTS which implements an attention based recurrent neural network (RNN) architecture, which combines text analysis and acoustic model components.

NN-based TTS systems have evolved rapidly since 2016 and have brought various advantages compared to earlier methods. The systems can be trained using large datasets of text and natural speech and the use of single model decreases the errors between multiple models and components (Ma et al. 2020).

Text analysis is a required component of TTS systems. The text input is normalized in a way that non-natural language symbols are converted into natural language (Xu et al. 2021). Symbols such as numbers, dates and currency are represented with words. For example, the number “123” is converted into “one hundred twenty-three”. Text and symbols can contain information about pronunciation. Grapheme-to-phoneme (G2P) conversion is used to translate characters into phonetic transcription to reflect actual pronunciation. For example, the word “play” is converted to “p l ei”. Phonemes are commonly used as input unit for acoustic models.

Acoustic model generates mel-spectrograms from the input units. The mel-spectrogram is spectral representation of audio signal, where frequency axis is converted into the Mel scale. There are different architectures for acoustic models which use various neural network methods. For example, Tacotron and Tacotron 2 systems are RNN-based, DeepVoice series system use convolutional neural networks (CNN) and FastSpeech

series uses Transformers to generate mel-spectrograms (Ping et al. 2017; Wang et al. 2017; Shen et al. 2018; Ren et al. 2019). The most recent TTS systems use large language model (LMM) based approach (Łajszczak et al. 2024). Each system can generate intelligible synthetic speech with comparable mean opinion scores (MOS). MOS score is a commonly used method to evaluate the quality and naturalness of synthetic speech.

CNN- and RNN-based models generate mel-spectrograms using autoregressive methods, which use previous elements to predict the next ones (Ren et al. 2019). TTS systems use large datasets to train models and require a lot of computational power. Non-autoregressive models can generate mel-spectrograms in parallel which provides much faster training and waveform generation (Ren et al. 2019; Xu et al. 2021). Therefore, many newer systems have implemented the non-autoregressive method (Miao et al. 2020; Ren et al. 2020; Łańcucki 2021). Non-autoregressive models have gained better results at MOS scores and faster performance compared to autoregressive models.

Vocoder is a model to convert mel-spectrograms, produced by acoustic model, into speech signal. Modern vocoders have applied neural network architecture and can generate high quality speech (Kong et al. 2020). In literature, commonly used vocoders are WaveNet (Oord et al. 2016), WaveGlow (Prenger et al. 2019) and HiFi-GAN (Kong et al. 2020). Each vocoder can produce high quality speech and to mention, HiFi-GAN outperforms earlier methods in synthesis speed and speech quality (Kong et al. 2020). HiFi-GAN uses CNN architecture to generate raw audio signal from mel-spectrograms non-autoregressively. Autoregressive vocoders share the same problem in performance as TTS systems. However, the autoregressive method is a simple way to take into account the long-term dependencies (Prenger et al. 2019). As the overall goal of TTS systems is to produce more natural-sounding speech, use of a different vocoder can make a difference.

2.2 Prosodic features of speech

Speech is a mean for interaction and communication. It is very efficient way to convey messages. Prosodic features refer to nonverbal features of speech and they play significant role in conveying messages (Frick 1985). Prosody has an impact on naturalness of speech as it delivers information that written words cannot. Speech can be described perceptually and acoustically (Frick 1985). Perceptual refers to the sense of hearing, which includes pitch, loudness, and duration. These can be measured and acoustically analyzed from the speech signal. The semantics of the phrase or sentence is dependent on the prosody. For example, emphasizing certain words by means of prosodic stress, highlighting certain syntactic phrasing by means of pausing, and differentiating general questions and statements by means of intonation. It is crucial for speech synthesis to take into account the prosodic features when generating natural and intelligible speech.

Fundamental frequency (F_0), also commonly called pitch, is measured in hertz (Hz) from the audio signal (Daniel & Céline 2022). F_0 refers to the lowest frequency of a periodic audio signal and is an acoustic property of speech. An example of F_0 contour for speech is presented in Figure 2.1 as a function of time. For voiceless phonemes and pauses there is no measurable values, which makes the curve discontinuous.

Mean pitch value for adult male speaker is in range 105 Hz to 125 Hz and for adult female speaker in 200 Hz to 220 Hz (Pisanski et al. 2021). The mean pitch is one the most prominent personal properties of speaker's voice. In general, different people have different pitch. Therefore, the same speaker's speech is used in the presented experiments both for TTS training and to represent the natural speech.

The perceived level of pitch differences is non-linear since pitch rise and fall are perceived smaller for high-pitched voice than low-pitched voice (Daniel & Céline 2022). To take into account the non-linear features and perceived frequency, the values can be converted to semitones. Semitones are the smallest difference in sound between two notes. The Mel scale is a perceptual scale of pitch, which is based on subjective listeners. It is widely used in speech processing and is a fundamental result of psychoacoustics (Umesh et al. 1999).

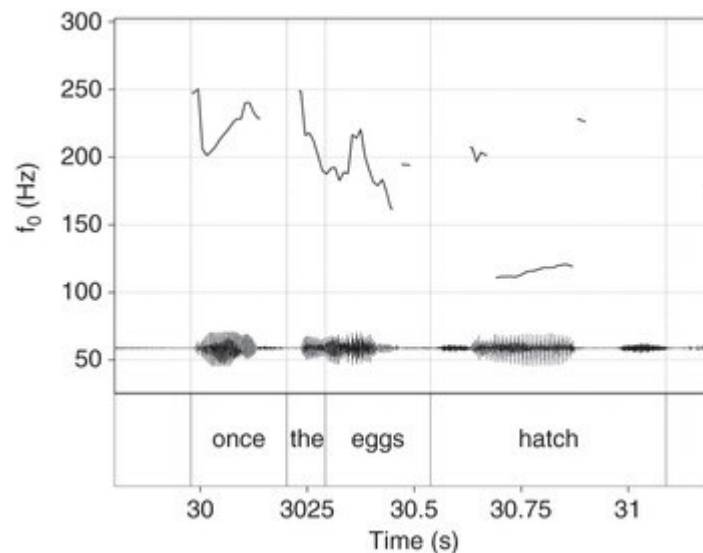


Figure 2.1. *F0 for phrase “once the eggs hatch” (Liberman 2017)*

Speech tempo varies in different situations and is a part of non-verbal communication. There are a lot of sources for tempo variation. Emotions, language proficiency and stress influence tempo (Breitenstein et al. 2001; Trouvain 2004). Social situations and interactions can also increase or decrease speech rate. For example, sports commentary commonly has high tempo. The speech tempo affects how humans perceive spoken language (Trouvain 2004) and therefore, it has influence on naturalness of the synthetic speech. Modern speech synthesis allows speech tempo manipulation, which can affect experiments and comparison between real and synthetic speech.

Speech rate can be measured using linguistic units such as word rate, syllable rate and phone rate. These measurements can be represented using units such as words per minute (wpm), syllables per second (s/sec) and phones per second (p/sec). The simplest measurements for word rate can be done by counting the words from the transcript. Automated speech recognition (ASR) can be used when the transcript of the speech recording is unknown. However, this method does not take into account the pauses in speech and possible word-error-rate (WER) can make it unreliable for comparison. Variability in word lengths can be problematic for expressing tempo since the lengths are correlated to duration. Smaller units are better for expressing temporal features (Trouvain 2004).

The intensity of speech signal is an important acoustic property. Loudness of speech is a perceived feature, whereas the levels of loudness are related to the spectral energy of sound (Seshadri & Yegnanarayana 2009). Spectral energy refers to different frequency

bands in audio signal. Signal energy is often measured with logarithmic unit decibel (dB). It is used to express the intensity or power of sound levels. However, energy can be measured using squared values from short segments, which represents the mean amplitude changes. Audio signal can be affected by speakers' distance to recording equipment. Signal may have higher or lower levels in amplitude depending on the distance between the recording device and speaker.

Loudness of speech is a perceived feature. Subglottal pressure refers to the air pressure, which is generated by lungs. Increase in subglottal pressure is related to loudness and pitch of speech (Frick 1985). Loudness contains information and can convey emotions and speaker attitudes. For example, raising the loudness of words and phrases can indicate emotions of speaker and lowering the loudness may signal uncertainty. In general, loudness varies between weak and loud voice. The energy distribution can be different between speakers since energy is influenced by different accent, speaking style and characteristics (Mary & Yegnanarayana 2008). Therefore, it is problematic to compare energy distributions between completely different speakers.

Spectral tilt represents the spectral characteristics of a speaker's voice. Spectral tilt measures can be done in different ways and study by Kakouros et al. (2018) shows difference methods for spectral tilt measures and comparison. Efficient and common method and used by Takeda et al. (2010) and Simantiraki et al. (2018) is the slope of the regression line, fitted to the dB-scaled short-term logarithmic magnitude spectrum.

Spectral features can provide information from the emotions of a speaker and time-frequency correlation (Jiang et al. 2019). Therefore, emotion recognition from speech have focused on spectral features (Takeda et al. 2010; Shukla et al. 2011; Simantiraki et al. 2018). Automatic emotion detection has wide range of applications from health care, robotics, and response systems in call-centers (Simantiraki et al. 2018). Takeda et al. (2010) study showed that spectral tilt measures for the emotions "anger" and "joy" are different than for neutral speech. For synthetic speech, it is important that correct emotion is transferred to the speech. The message could be understood wrongly if the emotion of the speaker differs from the text. Overall, the correct expression of emotions makes synthetic speech more intelligible and natural sounding.

2.3 Text-to-speech evaluation metrics

MOS is widely used to measure the perceived quality of synthetic speech. The MOS score has scale 1 – 5, where 1 indicates low score and 5 indicates excellent score. Score evaluations consist of a group of participants, which rates the speech using MOS scale. The participants can be asked to rate different features such as the overall quality, naturalness, and pronunciation. Usually, the MOS scores are evaluated before a TTS system is published. The use of MOS score provides the possibility compare the perceived quality of synthetic speech between TTS systems. However, MOS is vulnerable to misuse or misinterpretation (Streijl et al. 2016). Since the evaluators are human, there could be huge differences in MOS scores between two different evaluator groups. Moreover, Le Maguer et al. (2024) review shows that absolute category rating (ACR) does not produce enough information about the quality of synthetic speech. The ACR scale is a five-level rating, for example the output of a MOS test.

Objective quality tests are a great alternative for subjective tests. Objective methods offer less cost- and time-consuming tests. Whereas subjective tests take time as evaluators are gathered and tested, objective tests can be performed quickly and automatically. Commonly used objective quality metrics are mean squared error (MSE), word error rate (WER) and perceptual evaluation of speech quality (PESQ). MSE is a statistical method to evaluate the amount and significance of errors. TTS systems can use MSE for optimization, where MSE is evaluated between predicted and ground truth values (e.g mel-spectrograms or speech signals). For example, FastSpeech (Ren et al. 2019) model uses MSE loss to predict duration. WER is used to evaluate intelligibility of TTS output. WER is usually a Levenshtein distance between two sequences of words. The generated speech is recognized by ASR and its output is compared with pronounced text input.

PESQ is an objective metric to measure speech quality. PESQ score is calculated for audio signal using second signal as reference. The scores have scale of -0.5 - 4.5, where -0.5 indicates low quality and 4.5 indicates high quality. PESQ is mainly used to evaluate the quality of applications in telephone networks and speech codecs (Rix et al. 2001). However, it has been used as objective evaluator for synthetic speech (Cernak & Rusko 2005; Bhattacharjee et al. 2021).

The similarity of two populations can be examined with statistical analysis. The student's *t*-test is a statistical method to test if the difference between two populations is statistically significant or not. The student's *t*-test assumes that populations have equal

variances and sample size. Bernard Lewis Welch developed an alternative test, Welch's t -test, it is recommended to use when variances and sample sizes are unequal for the populations (Zimmerman & Zumbo 1993).

The distributions are compared by statistical test, which is aimed to answer whether distributions are different and what is a numerical difference between them. Student's t -test estimates the significant differences, but evaluating the distance between distributions can provide more information on how different the distributions are. The Wasserstein metric, presented in section 2.1, is a method for measuring the distance between distributions. The Jensen Shannon (JS) Divergence is a metric for similarity between two probability distributions. In literature, it has applications in audio summarization (González-Gallardo et al. 2019) and prosodic distributions (Li et al. 2023). The JS Divergence scores are non-negative and values closer to zero indicate identical distributions.

It is important to note that recording equipment and environment has an effect on quality. Background noises can affect both the quality of the trained TTS model and comparison results. Therefore, usually the training data for TTS systems consists of a high-quality speech with high signal-to-noise ratio (SNR). SNR is a measure of the ratio between speech signal and background noise. Although the latest TTS systems aimed to train on several 100K hours of speech do not require high quality speech data.

3. RELATED WORK

Speech synthesis has brought together linguistics, communication, and machines. Linguistics is the study of language, which has branches such as phonetics. Phonetics focuses on speech and sound production. However, communication includes more information than words can deliver. Non-verbal communication contains changes in physical behavior but also changes in speech production such as tempo, style, and pitch. Prosodic features are used for language identification and classification. Prosodic features can be studied from the speech audio signal. Typically, research focuses on features such as pitch, tempo, and energy (Mary & Yegnanarayana 2008; Minixhofer et al. 2022; Li et al. 2023;).

3.1 Evaluating and reducing the distance between distributions

Minixhofer et al. 2022 introduced a method to measure the distance between real and synthetic speech distributions. The study used a metric approximation to evaluate distances of measurements of acoustic environments, speakers, and prosodic features. Environmental measurements were computed using Speech-to-Reverberation Modulation energy Ratio (SRMR) and Waveform Amplitude Distribution Analysis (WADA) to represent the reverberation and noise in the audio signal. Measurements for speaker characteristics were computed using the Fréchet distance (FD). Measured prosodic features were pitch, energy, and duration. Distance between real and synthetic distributions is computed measuring the 2-Wasserstein distance for cumulative distribution functions. Results show that prosody and speaker distributions can be reduced for the FastSpeech 2 (Ren et al. 2020) TTS model.

3.2 Diverse and expressive speech prosody prediction

Li et al. 2023 research predicts prosody with denoising diffusion probabilistic model. The proposed prosody predictor is used with FastSpeech 2 (Ren et al. 2020) TTS

model. The FastSpeech 2 is trained with Mandarin audiobook dataset and the expressiveness of the synthetic speech is evaluated with a MOS test. The difference between prosody distributions of the prediction and ground truth is evaluated with Jensen-Shannon Divergence. Measured prosodic features are pitch, energy, and duration.

The computed JS Divergence values are lower for every feature. For energy, the difference is a minimal change in value, with unit of one thousandth. For the pitch and the duration, the values are approximately half of the original model. The JS Divergence values are also represented using histograms, which shows values for each phoneme. The JS Divergence shows reliable results, which is evidenced by comparison and correlation of the JS Divergence values and figures of the prosody distributions. Overall, the research and experiments show that proposed model can predict prosody with greater diversity. (Li et al. 2023.)

4. EXPERIMENTS

4.1 Experimental data

Experimental data for the current research consists mainly of the LJSpeech dataset, which is a large public domain speech dataset (Ito 2017). It contains approximately 24 hours of short audio clips from a single speaker reading from seven different audiobooks. Length of audio clips vary between 1 to 10 seconds. The LJSpeech dataset also contains transcripts for audio clips. The transcripts are normalized, meaning the numbers, monetary units and abbreviations have been expanded into full words and for example, the abbreviation “Mr.” is expanded into the word “Mister”.

The speaker in the LJSpeech dataset is Linda Johnson. Recordings are from LibriVox, which is a non-profit project for free public domain audiobooks. The texts are matched to the audio manually, which provides reliable use. LJSpeech is a state-of-the-art dataset to train and compare various TTS systems.

Since the FastPitch TTS model and the HiFi-GAN vocoder are trained using the LJSpeech dataset, the comparison in this study is made with Linda Johnson’s natural speech. For natural speech, we have collected audiobook recordings by the same speaker from LibriVox which are not included into LJSpeech corpus. Recordings are chapters from two different books, *From Pillar to Post* (Bangs 1916) and *Prejudices, First Series* (Mencken 1919). Overall, we have collected six chapters, and the length varies between 16 to 32 minutes and approximately 2.5 hours in total. Further on, the experiments use the term “*audiobook*” for the chapters *American magazine*, *Emergencies*, *Good Samaritan*, *Perils of the platform*, *Potrait of an immortal soul* and *Ulster polonius*.

The transcripts for the collected natural speech are preprocessed and normalized using tools from the Nemo Framework text processing Python package (Bakhturina et al. 2022). The preprocessing converts the transcript to a list of sentences which are normalized in the same way as the LJSpeech dataset. Additionally, the invalid character “æ” was manually changed into characters “ae” and three- and four-dot ellipses were removed.

4.2 FastPitch

FastPitch TTS model was used within the current research. FastPitch provides robust and high-quality synthetic speech. NVIDIA NeMo Framework has its implementation and pretrained models for it (Harper et al. 2022). Mel-spectrograms were generated with pretrained FastPitch model and synthesized to waveform using a pretrained HiFi-GAN vocoder model. Both models were trained with LJSpeech dataset.

FastPitch TTS model is based on FastSpeech architecture (Łańcucki 2021). FastPitch model focuses on explicit prediction of pitch contour and speech segment durations for the generated speech. Generated synthetic speech has high quality and natural sounding pitch. Sound quality is state-of-the-art with an MOS-score of 4.08 when synthesizing with WaveGlow vocoder (Łańcucki 2021). For comparison, with the same vocoder, Tacotron 2 has a score of 3.95 and FastSpeech 3.84. HiFi-Gan vocoder has received better MOS scores compared to WaveGlow.

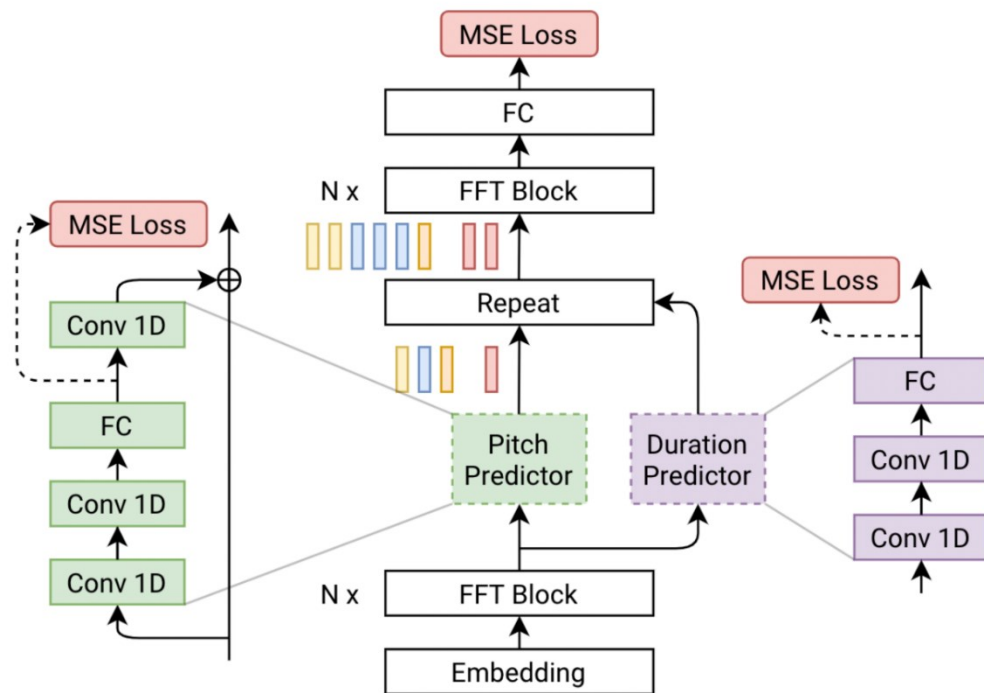


Figure 4.1. Architecture of FastPitch (Łańcucki 2021).

Architecture of FastPitch is shown in Figure 4.1. It is mainly constructed on two feed-forward Transformer (FFT) stacks. Between stacks, the predictions of pitch and duration are made with hidden representation of input characters. The pitch and duration values are predicted for all characters using stacks of 1-Dimensional CNN layers. The pitch values are embedded and added to character representation to shift them, and the result is transferred to output Transformer stack, which generates the mel-spectrogram. For training, the system uses ground truth values of pitch. The general loss function is a combination of MSE loss for pitch, durations, and mel-spectrogram.

It is difficult to generate synthetic speech for long text sequences to waveform due to limited amount of GPU memory. Therefore, the synthesized audio was generated by sum of multiple short waveforms. Length of waveform was dependent on utterances in sentence and a speaking rate. A 0.4 second pause was added between waveforms to represent breaths between sentences.

Nemo framework provides possibility for pitch and speech rate manipulation for synthesized speech. These parameters are not altered, and speech is synthesized with default parameter values.

4.3 Prosodic features of natural and synthetic speech

The experiments include comparison for prosodic features. Features such as F0, speech rate, spectral tilt and energy are measured for each audiobook presented in section 4.1. Same features are measured for synthesized audiobooks, which are generated using the presented method in section 4.2. Features are extracted identically for both real and synthetic audiobooks. Python packages Librosa and Parselmouth were used for feature extraction (McFee et al. 2015; Jadoul et al. 2018). Librosa is a tool for audio and music processing. Parselmouth is Python interface for Praat software, which is software for speech analysis in phonetics, designed and developed by Paul Boersma and David Weenik.

As for F0, we make the comparison between F0 values and do not convert values into logarithmic representation. F0 values are extracted by Parselmouth using autocorrelation method (Boersma 1993). We use floor value of 75 Hz and ceiling value 600 Hz and 40 ms window length.

Speech rate distribution was estimated as number of syllables, i.e vowels, per second. The syllable rate was measured using a Praat script, designed by De Jong & Wempe

(2009). Vowels within a syllable tend to have higher energy and the algorithm detects syllables from the audio signal by examining peaks in the energy and intensity contours. Values are measured using 4 second window length and 2 second hop length. The amount of the syllables is then divided by the window length to get the syllables per second unit.

Spectral tilt is measured by fitting a slope to the short-term logarithmic magnitude spectrum. Short-time Fourier transform is computed using Librosa. We use sampling rate of 22050 Hz and window length of 2048 samples, which is approximately 93 ms. Hop length is half of the window length, i.e. 1024.

Speech energy is measured using root-mean-square (RMS) energy. RMS energy is calculated using following formula:

$$\sqrt{\frac{1}{N} \sum_n x(n)^2}.$$

We use window length of 1024 samples and hop length 512. Zero values are excluded. Librosa has a method for computing RMS values, but results are the same as following the formula, while computing time is slightly slower.

PESQ scores are measures using the variation MOS-LQO (Listening Quality Objective), which results are mapped to scale 1.0 - 4.5. The results are more comparable with MOS score. We split the real and synthetic audiobooks into equal number of segments for calculations to minimize the impact of the variability between signal lengths. Cernak & Rusko (2005) found that PESQ could not be used to evaluate quality for short clips of speech. With 50 equally sized segments, the length varies approximately between 19 and 38 seconds and includes respective number of words. Calculations are made with the Python PESQ package, developed by Wang et al. (2022).

Statistical analysis is computed using Welch's *t*-test. For deeper statistical analysis, we compute Jensen-Shannon Divergence. Scores for *t*-test and JS Divergence are computed using Python Scipy package.

4.4 Results and discussion

Prosodic feature extraction and statistical analysis are conducted for the audiobooks presented in section 4.1. The distributions of prosodic features are shown for each audiobook in Figures 4.3, 4.4, 4.5 and 4.6. The representations are in violin plot format, which draws the numeric data using a kernel density estimation (KDE). The width of the curve corresponds to quantity of the data points. The distribution for real speech is shown in blue color and for synthetic speech in orange. The quartile lines are added for better visualization.

Statistical analysis was conducted to state the differences between distributions of prosodic features of real and synthetic speech. Welch's t -test was chosen for testing method over Student's t -test since sample sizes are unequal due to speech rate differences. Results show that there are significant differences for every feature between real and synthetic speech.

To analyze differences between distributions, we use JS Divergence method to measure the similarity. Values of JS Divergence is presented in Table 4.1. Values indicate small differences for features RMS energy (RMSE) and spectral tilt (ST). Highest differences are for speech rate (SR) distributions for audiobooks American magazine and Ulster polonius. The higher differences can be seen as greater difference between mean lines, when observing Figure 4.3.

Table 4.1 Jensen-Shannon Divergence values for each audiobook and feature.

Audiobook	Features			
	F0	SR	RMSE	ST
American magazine	0,27	0,47	0,18	0,12
Emergencies	0,21	0,25	0,08	0,08
Good samaritan	0,25	0,21	0,14	0,11
Perils of the platform	0,19	0,14	0,07	0,08
Portrait of an immortal soul	0,23	0,37	0,14	0,11
Ulster polonius	0,26	0,46	0,16	0,11
Mean	0,24	0,32	0,13	0,10

PESQ scores for each audiobook are seemingly low. Results for PESQ scores are presented in Table 4.2. With the mean value of 2.02 for all audiobooks, synthetic speech should be hardly understood with bad quality. Ground truth values for PESQ are 4.64. With deeper analysis, a discovery was made that majority of the scores are between 1 and 1.5. The distribution of PESQ scores for American magazine audiobook is presented in Figure 4.2. If we remove the values below 1.5, the mean PESQ score for American magazine would be 2.93, which indicate slightly better quality.

Table 4.2 PESQ scores for each audiobook.

Audiobook	PESQ
American magazine	2,28
Emergencies	2,02
Good samaritan	1,67
Perils of the platform	1,85
Portrait of an immortal soul	2,27
Ulster polonius	2,02
Mean	2,02

We listened to the segments with low PESQ score and the speech is more natural sounding than what the scores represent. Cernak & Rusko (2005) observed that PESQ scores of short segments (10 s) cannot be used for evaluation, our results show that longer segments (17 – 38 s) does not evaluate the quality well either. There are no significant differences between the PESQ scores for audiobooks, i.e. the duration of segment does not affect the PESQ score.

Since the PESQ scores are calculated using the real speech audio signal as reference, the slight differences in audio signals can have influence on scores. The synthetic speech does not have outliers, which makes slight differences on the audio signal. The PESQ score was calculated by dividing the audio signal to approximately same speech segments, which can also be reason for larger differences in certain segments.

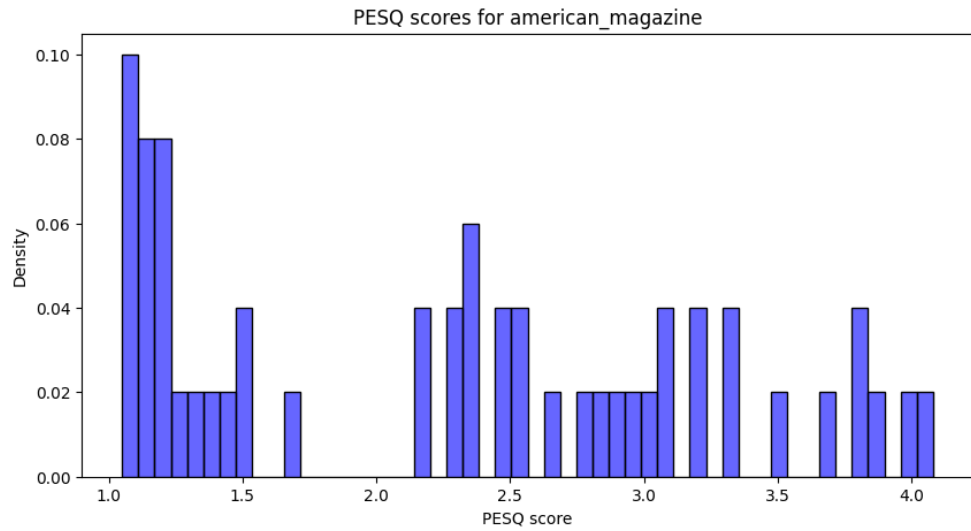


Figure 4.2. PESQ score distribution for audiobook *American magazine*.

The distributions of F0 values are shown in Figure 4.3. For real speech, there are differences between audiobooks as half of the KDE curves have mode below 200 Hz, while the other half has more variability.

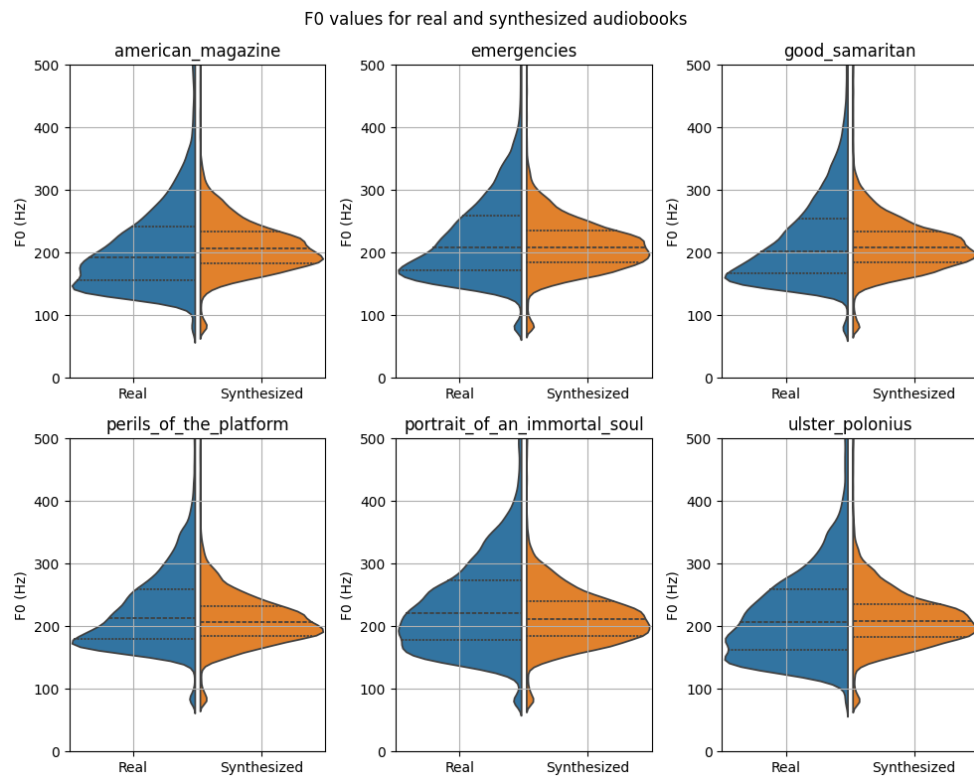


Figure 4.3. F0 distribution for each audiobook.

For synthesized speech, the KDE curves are more similar, and modes have slight differences. The median values for synthetic speech are slightly higher and more stable than for real speech. Mean F0 value for real speech distributions is 222 Hz and 213 Hz for synthetic speech.

Differences in the F0 values between audiobooks for real speech show that how the speaker's pitch changes throughout different audiobooks with different context. However, the synthetic speech does not have outliers and highest values are the same for each audiobook, approximately 330 Hz. Real speech has more variability in F0 values, especially in higher Hz values.

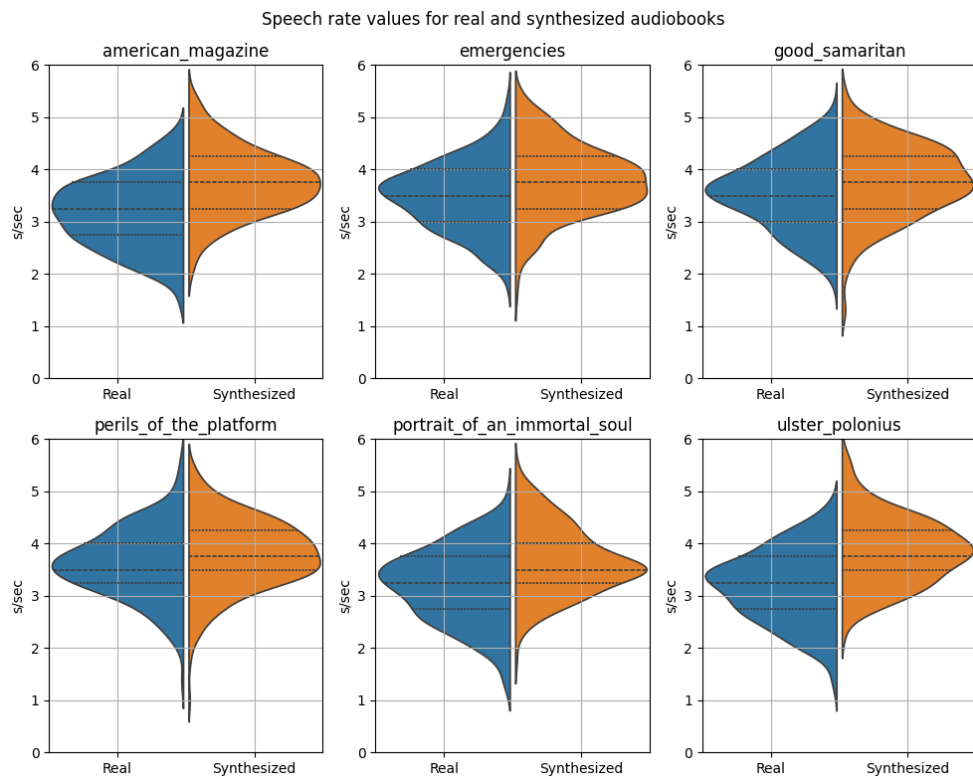


Figure 4.4. Speech rate distribution for each audiobook.

The distributions of speech rate values are shown in Figure 4.4 as syllables per second. For real speech, distributions and the mode occur in the same value, approximately 3.5 s/sec. The modes and median lines are between 3 to 3.5 s/sec and mean value for all recordings 3.37 s/sec. Synthesized speech modes and median lines are between 3.5 to 4 s/sec and mean value for all recordings 3.78 s/sec. Synthetic speech curves vary between different audiobooks.

Overall, the synthetic speech has a greater speech rate. The synthesized audiobook was constructed by adding short, synthesized sentences together and each sentence synthesized sentence has lower duration. The algorithm used to measure speech rate can extract the manually included pauses and they do not affect the results. The impact of faster speech rate is more visible for long recordings as the total length of synthesized audiobooks is approximately three to four minutes shorter. The difference is smaller for shorter audio clips. Speaker tempo can vary due to different reasons and audiobooks are all different recordings. Each audiobook has a different story and context, which makes natural differences for speech tempo.

FastPitch TTS system was trained using short segments of speech. Durations of LJSpeech are between 1 and 10 seconds (Ito 2017). This could affect experiment results as the synthesized segments are partly longer. However, the subjective listening evaluation for American magazine audiobook showed, that longer segments sound natural. Longer segments could explain results for faster speech rate of synthetic speech if the TTS system predicts the duration to fit more in the window of trained speech segments.

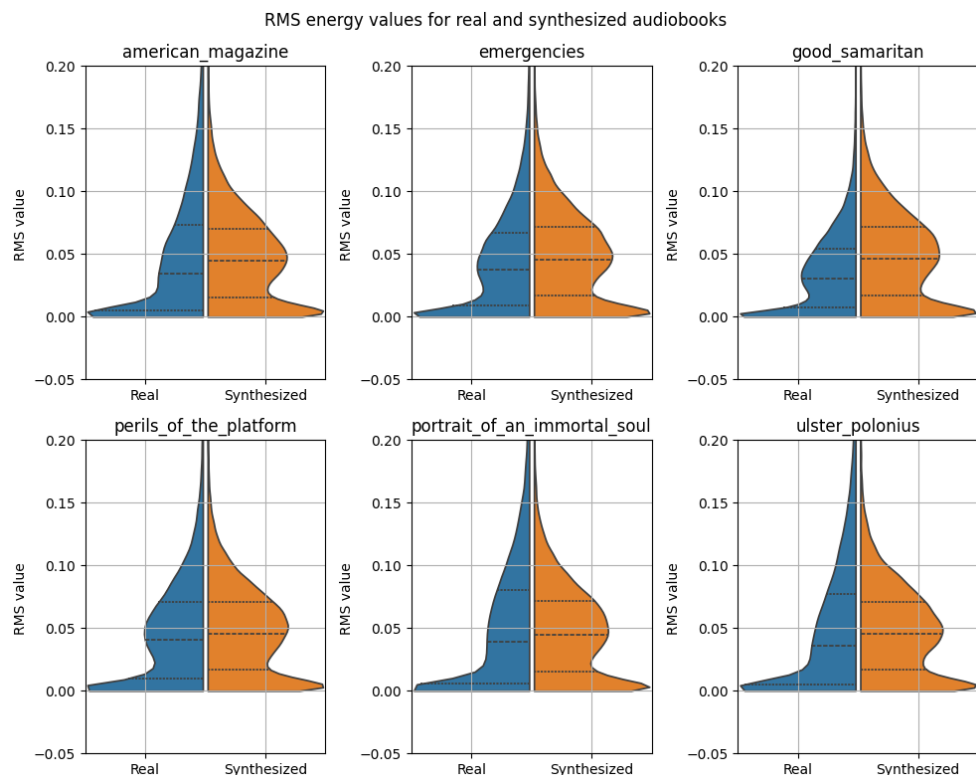


Figure 4.5. RMS energy distribution for each audiobook.

Energy distributions are shown in Figure 4.5. The RMS energy values represent root-mean-squared energy over segments of time. Zero values represent the silent segments in recordings. Higher values correspond to higher intensity or loudness. The median line of synthetic speech is almost the same for all audiobooks. Real speech has differences between audio recordings, half of the recordings do not have a mode at level 0.05 and energy distribution is linear when excluding zero data points. Mean values for all recordings are 0.046 for real speech and 0.048 for synthetic speech.

The TTS system can generate a wide range of energy levels. However, synthetic speech does not have higher levels of energy (0.20) which real speech has for the audiobooks *Potrait of an immortal soul* and *Ulster polonius* in Figure 4.5. Overall, the synthetic speech has similar curves and there are no significant differences between recordings. The differences between real speech recordings may be affected by external matters such as the distance of the speaker from recording device or different parameters for different recording sessions.

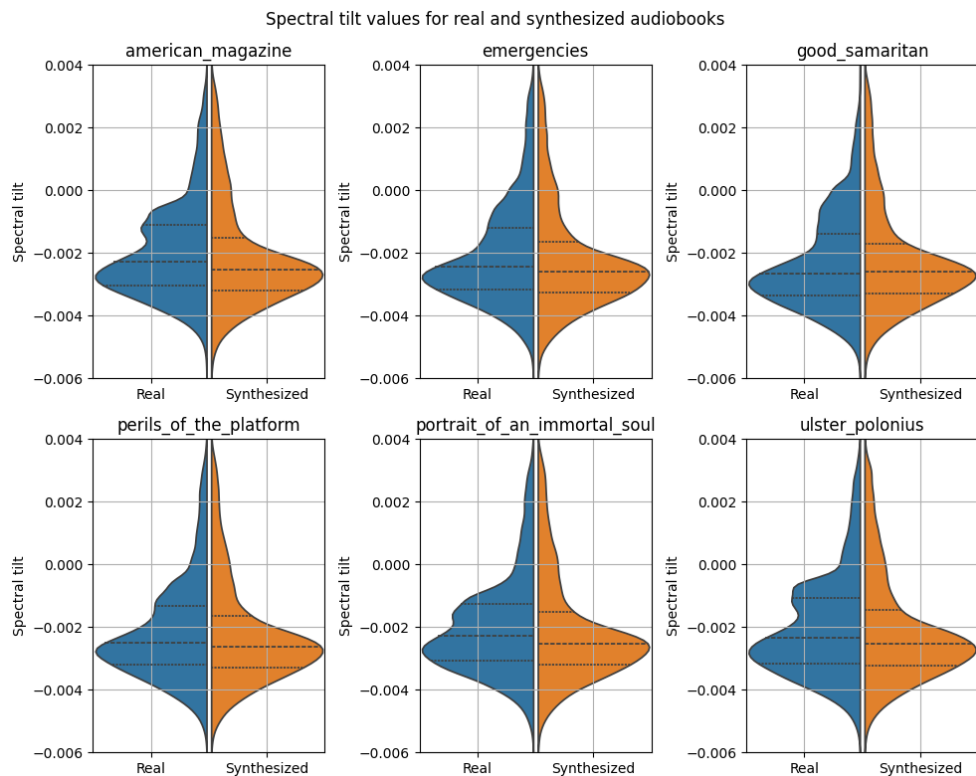


Figure 4.6. Spectral tilt distribution for each audiobook.

Spectral tilt distributions are shown in Figure 4.6. Values on the y-axis represent the slope of the fitted first order polynomial line. Fitted lines get negative and positive val-

ues and have small slopes. Synthetic speech does not have significant differences between audiobooks. Real speech has a small variability between values -0.002 and 0. For both real and synthetic speech, the mode of KDE curve is in the same place. However, synthetic speech seems to be unable to generate the second mode, which occurs for the real speech. Mean values for all recordings are -0.002091 for real speech and -0.002173 for synthetic speech. Values are similar considering the least significant number in distributions.

Statistically the difference between spectral tilt values was the lowest. The audiobook recordings have less emotional variation than spontaneously and naturally occurring emotions. The TTS system was trained using the audiobook recordings and speaker spectral features are assumed to be the same. The lowest difference in JS Divergence seems reasonable for the feature spectral tilt since the audiobook recordings have less emotional variation.

5. CONCLUSIONS

The goal of this thesis was to evaluate how well synthetic speech inherits prosodic features of the natural speech it is trained on. We successfully synthesized the speech with high quality. The comparison of prosodic feature distribution is in section 4.3.

The experiments showed that there is a difference between prosodic distributions for real and synthetic speech. Overall, synthetic distributions had slight differences between recordings for each feature, which implicates the contextual differences between book transcripts. The large statistical difference was found for F0 and speech rate. The least statistical different features are energy and spectral tilt. The results may be explained by the fact that F0 and speech rate variability highly depends on linguistic meaning of the text, while spectral tilt is an intrinsic property of speaker's voice. As for energy, RMS values were averaged over time and might not capture the intensity variations. Other methods could be used to analyze these variations more accurately.

The FastPitch (Łańcucki 2021) TTS system uses pitch and duration predictions for the generation of mel-spectrogram. The experiments had the biggest differences for F0 and speech tempo, which are influenced by the pitch and duration predictions. This observation indicates that FastPitch still struggles with modelling correct pitch and durations. However, the synthetic speech has high quality and intelligibility, which indicate that predictions are in a good level. Future work should include comparison for specially selected lists of sentences to test potential limitations for generation of proper prosodic features.

The experiments should be done for more spontaneous speech to evaluate more naturally occurred prosodic variations. For example, podcasts where there are no pre-written transcript and one or multiple speakers' emotions and speech have natural variations. However, these experiments would require a large dataset with transcripts.

Further work should consider other features and methods for extraction. Evaluations should be made for newer and future speech synthesis systems. As the quality of synthetic speech increases, the speaker identification systems need to advance. Applications which use speaker identification are exposed for misuse. However, with better speech quality, the applications would become more effective and human friendly. For example, applications in education for different languages and translators for general use.

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